

Binary Payment Schemes: Moral Hazard and Loss Aversion

Herweg, Müller, and Weinschenk 2010
A Literature Review

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Outline

- 1 Contract Theory & Incentive Schemes
- 2 Theoretical Framework
- 3 Model Results
- 4 Empirical Support
- 5 Outlook

Contractual Form Matters: A Problem of Decades

- **Observation:** Real-world contracts are often remarkably **simple** (e.g., binary bonus schemes, flat wages).
- **Classical Contract Theory** (Holmström 1979): Optimal contracts should be **complex**, exploiting all available information.

The Central Puzzle

If contractual form matters so much, why do we observe such a prevalence of relatively simple contracts?

The Puzzle of Contract Simplicity

Prendergast (1999): *The Provision of Incentives in Firms*

- **Established the puzzle:**¹ Real contracts are far simpler than theory predicts.
 - ▶ Highlights the disconnect between complex theoretical prescriptions and standard contractual practice.
 - ▶ **Methodological limit:** Survey-based, no causal identification of *why* simplicity prevails.

Lazear and Oyer (2007): *Personnel Economics*

- **Sharpens the puzzle:** Argues there must be benefits to simple contracts that classical theory misses.
 - ▶ Bridges the gap between theoretical models and applied personnel economics.
 - ▶ **Open question:** What *exactly* are these benefits? — This is where Herweg, Müller, and Weinschenk (2010) enters.

¹Baker (1988) first documented this discrepancy

A Behavioural Explanation

Herweg, Müller, and Weinschenk (2010) address this puzzle by introducing **expectation-based loss aversion** into the standard principal–agent model with moral hazard, building on Köszegi and Rabin (2006).

The Köszegi–Rabin Concept:

- Individuals evaluate outcomes relative to a **reference point**; losses loom larger than comparable gains (Kahneman and Tversky 1979).
- In stochastic settings, the reference point is the **entire probability distribution** of wages the agent anticipates.
- The agent compares the realised wage against **every possible wage realisation** in that reference lottery, incurring a psychological cost for each unfavourable comparison.

Main Result: Why Binary Schemes?

In Herweg, Müller, and Weinschenk (2010), the agent plays a **Choice-Acclimating Personal Equilibrium (CPE)**: he correctly anticipates the wage distribution induced by his effort, and this anticipated distribution becomes his reference point.²

Herweg, Müller, and Weinschenk (2010) identify a trade-off of two opposing effects of adding wage levels:

- 1 **Incentive Gain:** More wage levels (complexity) provide better incentives. More wage levels $\Rightarrow$ finer performance signals $\Rightarrow$ higher marginal returns to effort.
- 2 **Loss Aversion Cost:** Every additional wage level enlarges the support of the agent's stochastic reference distribution. This creates more possible unfavourable comparisons (realised wage falls short of other anticipated outcomes), raising expected psychological costs.

²See Köszegi and Rabin (2007) for the formal definition of CPE.

Do Expectations Really Matter? Empirical Support

The behavioural foundations of the model by Herweg, Müller, and Weinschenk (2010) rest on a well-established empirical foundation:

- **Labour Supply (Cab Drivers):** Crawford and Meng (2011) shows that a model with expectation-based targets significantly outperforms the neoclassical benchmark in predicting cab drivers' effort.
- **Lab Evidence (Real-Effort):** Abeler et al. (2011) demonstrates that directly manipulating subjects' expectations shifts effort provision in the direction predicted by loss aversion models.

From Theory to Experiment: Looking Ahead

Why Binary Schemes?

The central prediction, that binary schemes are optimal, has not yet been directly tested. This is addressed in the second presentation.

Subsequent Theoretical Work: Herweg and Schmidt (2015) extend the framework to dynamic settings:

- Expectation-based loss aversion can explain *inefficient renegotiation*
- Evidence that the mechanism has explanatory power beyond **static** moral hazard

Backup: A Binary Scheme

A binary scheme contains only two wage realisations, minimising loss aversion costs while still providing meaningful incentives.

The Optimality Theorem

When loss aversion is sufficiently high, the principal minimises these psychological costs by offering a **base wage** and a **single bonus**, even if a much richer set of performance signals is available.

Standard contract theory instead predicts a *fully contingent* wage schedule (Holmström 1979).

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